

MAVRA middleware

1-5 use

6-8 execution

learning

→ call / response

→ shared-store write

---> database read



Analysis optimizer

LLM agent: learns definitions; admission checks them

publish

Request handling

Lookup

match name, compare versions

Same-snapshot validation

check on the query's snapshot

Shared store

Definitions

return amount
pending

return rate
valid

store revenue
valid

Check results

per condition and table version, shared by all definitions and agents

one row per return
store_returns 15 ✓

one date per date key
date_dim 3 ✓

no lost dates
store_returns 15 ?

3 missing result: run check

4 store result

Maintenance

run missing checks · repair or invalidate

check queries, table versions

run the query on the snapshot

User agents



Return amount
in May?

1 "return amount"

5 valid definition

6 SQL query

8 query result

SQL

declares revision 2

ETL and writers



updates

SQL database

snapshot



store_returns
date_dim

14 → 15
3

store_returns

date_dim